Final Revision — Written Against Your Test

Diagnostic score **6 / 56**. Every topic below is ordered by what the test says you are missing, and every fact is sourced from your **class notes first**, the lecture slides second, and J. M. Smith third — because the quiz follows the class, not the book.

Attempted 9.8 min of 60 **Strongest** physisorption vs chemisorption (50%)

Zero-scoring Langmuir, densities, pore distribution, BET numericals

READ THIS FIRST 2 MINUTES

What the test actually measured

You stopped at 9.8 minutes and wrote “no idea” on eleven items. So this is **not** a measure of what you can do — it is a measure of what is **available to you without notes**, which right now is close to nothing. That is a recall problem, not an ability problem: on C2 you said the Langmuir fit is easy with calculator regression, and it is. The fix is short and mechanical.

Topic	Score	What the test showed
Physisorption vs chemisorption	2 / 4	Best answer on the paper. Bonds, energy, reversibility — all correct. Missing: temperature, specificity, and which one measures what.
Global vs intrinsic, seven steps	1.5 / 4	“3 to 5 are intrinsic, rest are mass transfer” is exactly right. You just could not list the seven.
External area / geometry	1.5 / 4	You recalled the shape `6/(ρd)` but botched units and inverted part (c).
Classification, preparation, promoters	0.5 / 5	“No idea” — yet this is four full pages of your own class notes.
BET	0.5 / 8	You have $v_m = 1/(I+s)$ and nothing else. No c, no α, no S_g route.
Langmuir isotherm	0 / 8	Biggest single loss. It is a six-line derivation you have in your notes.
Densities and void volume	0 / 9	Largest block of marks on the paper, completely blank.
Pore-volume distribution	0 / 8	You guessed pore sizes as “<5 and >20”. Right instinct, wrong numbers — see §4.
Global rate derivation, three regimes	0 / 3	Four algebra lines. Cheapest marks you can buy.
External transport + thermal effects	0 / 3	Book-only material, lowest priority of everything here.

ONE GENUINELY GOOD SIGN

Your confidence flags were **honestly set**. Everything you marked “Shaky” was in fact shaky, and the one thing you marked “Sure” (A1, global rate) was right. You are not fooling yourself about what you know — that is the hard half of revising, and you already have it.

IMPORTANT — A GAP IN WHAT YOU WERE STUDYING

The study material you have been revising from is built from J. M. Smith. Your **class notes contain five topics that are not in it at all**: the Temkin and Freundlich isotherms, the IUPAC Type I–V isotherm shapes, the IUPAC pore-size convention (2 nm / 50 nm, *not* Smith's 100 Å), catalyst shapes with the ΔP and S_A/V trade-off, and the five-way density vocabulary from the slides. The quiz is set by the person who taught the class. **§2 and §4 below carry that material** — treat it as the highest-yield reading here.

Fix these eight first

Ordered by marks recovered per minute spent. If you only get through this page, you move from 6/56 to roughly 30/56.

1. **The Langmuir derivation.** Six lines, ends in a boxed result. Worth 8 marks on my paper and it is Lecture 12 material. §3 below.
2. **He–Hg densities.** Two divisions and a subtraction; 9 marks. Helium → ρ_s, mercury → ρ_p, difference → V_g. §5.
3. **BET: from plot to S_g.** You already have $v_m = 1/(I+s)$. Add three more facts — $c = s/I + 1$, $\alpha = 16.2$ Å², and $S_g = (N_0 v_m / V_{STP})\alpha$ — and 8 marks open up. §3.
4. **The seven steps, listed.** You know 3–5 are the chemical core. Learn the list in order; it is one slide of your lecture notes. §1.
5. **Pore-size numbers.** Micro < 2 nm, meso 2–50 nm, macro > 50 nm (IUPAC, your notes). Micropores are inside the particle and hold the area. §4.
6. **Catalyst types and preparation.** Metal / semiconductor / insulator, and the precipitation → ... → activation flowsheet. Straight recall, 5 marks, four pages of your own notes. §6.
7. **Resistances in series.** Four algebra lines from two rate expressions to $1/k + 1/k_m$. §1.
8. **The $6/(\rho_p d_p)$ unit drill.** You had the formula and lost the marks on centimetres vs metres. §7.

Rates, the seven steps, resistances in series

1.1 Three ways to write a rate

From your notes — the same reaction, three normalisations, and they must agree:

$$-r_V = (1/V)(dN/dt) \quad mol/(m^3{\cdot}s) \quad \cdot \quad -r_W = (1/W)(dN/dt) \quad mol/(kg{\cdot}s) \quad \cdot \quad -r_S = (1/S_A)(dN/dt) \quad mol/(m^2{\cdot}s)$$

class

$$(-r_V)V = (-r_W)W = (-r_S)S_A$$

the bridge

Global vs intrinsic. The **intrinsic** rate is the surface chemistry alone — steps 3, 4, 5. The **global** rate is all seven lumped, written in **bulk** conditions, because bulk temperature and concentration are the only things you can measure or specify. Crushing the pellet changes the **global** rate only: it shortens the diffusion path. That was your A1 answer — the mark was for the *reason*, so always add it.

1.2 The seven steps — learn the list

#	Step	P/C	Note
1	External (interphase) transport of A, bulk → outer surface	P	Vanish for a **nonporous** catalyst: steps 2 and 6
2	Intraparticle diffusion of A, pore mouth → active site	P	
3	**Adsorption** of A on the active site	C	These three **are** the intrinsic rate
4	**Surface reaction** A → B	C	
5	**Desorption** of B	C	
6	Intraparticle diffusion of B, site → pore mouth	P	
7	External transport of B, surface → bulk	P	

Mnemonic that matches your own sentence: **transport in, diffuse in, A·S·D, diffuse out, transport out** — the middle three are chemistry, the outer four are physics. Each step has its own activation energy, and $E_A \propto 1/rate$ in the sense that the slowest step governs.

1.3 Resistances in series — the four-line derivation

Steady state means the two rates are equal, so eliminate the surface concentration you cannot measure:

$$r_p = k_m a_m (C_b - C_s) \quad = \quad k\ a_m\ C_s \qquad (7\text{-}1),\ (7\text{-}2)$$

$$\Rightarrow C_s = k_m C_b / (k_m + k) \qquad (7\text{-}3)$$

$$\Rightarrow r_p = a_m C_b / (1/k + 1/k_m) \qquad (7\text{-}4) \text{ — the one to memorise}$$

Read it as electrical resistances: **1/k is the chemical resistance, $1/k_m$ the transport resistance, and they add**. Three regimes (Fig. 7-1):

Regime	Condition	Profile from bulk to surface	Global rate
Reaction control	$k << k_m$	Flat — $C_s \approx C_b$	$r = k\,a_m C_b$
Diffusion control	$k_m << k$	Steep fall to $C_s \to 0$	$r = k_m a_m C_b$
Intermediate	$k \approx k_m$	Sloping, $0 < C_s < C_b$	full Eq. (7-4)

ONLY IF YOU HAVE SPARE TIME — BOOK-ONLY, NOT IN YOUR CLASS NOTES

Why raising gas velocity can lower an exothermic rate. Faster gas raises k_m (helps: C_s rises toward C_b) but also raises the heat-transfer coefficient, cooling the surface back toward bulk temperature (hurts: for an exothermic reaction a cooler surface is a slower one). Net direction depends on which dominates. Smith's cases: SO_2 on Pt is mass-transfer dominated (ignoring the concentration drop overstates the rate ~350%); H_2 oxidation on Pt/alumina is thermally dominated (ignoring ΔT is 20% low); UO_2 hydrofluorination is thermal-wins (+38% net).

Catalyst shapes, and why shape is a trade-off

Two full pages of your notes, zero of it in the material you revised from. Cheap marks if it is asked, and the Tutorial 1 style (commercial names, shapes, reactors) says it may well be.

Shape	Pros	Cons	Used in
Spherical	Low manufacturing cost	High pressure drop; large diffusion length	HDS, methanation
Irregular granule	Low manufacturing cost	High ΔP; low surface area	Ammonia, formaldehyde
Pellet (cylinder)	Regular shape, good strength	Low surface area	CO shift, hydrogenation
Extruded cylinder	Low pressure drop	Poor shape and strength	HDS
Monolith	Very low ΔP; small diffusion length	Poor radial heat transfer	Exhaust-gas cleaning
Wagon wheel, trilobe, hollow extrudate	High S_A/V at moderate ΔP	Weaker than solid shapes	the usual compromise

The three curves in your notes are the whole argument:

- **S_A/V against ℓ/d_p** — sphere and disc sit low; monolith, wagon wheel, hollow extrudate sit high. More shaping = more area per volume.
- **ΔP against d_p** — Ergun: $\Delta P = 150\mu v(1-\varepsilon)^2/(d_p^2\varepsilon^3) + 1.75\rho v^2(1-\varepsilon)/(d_p\varepsilon^3)$. Small particles give area but choke the bed; $\Delta P \propto 1/d_p$. If ΔP rises you need a bigger compressor — that is the real cost.
- **Strength against S_A/V** — falls as you shape it. So there is a **minimum acceptable strength** and a **minimum acceptable rate**, and the shape you pick lives between them.

One line worth memorising: **surface area, pressure drop and mechanical strength cannot all be maximised — catalyst shape is the compromise between them.**

Adsorption, Langmuir, BET

3.1 Physisorption vs chemisorption — your best topic, finish it

	Physical adsorption	Chemisorption
Force	Van der Waals	Chemical bond
Specificity	None — any surface, all of it	Specific — chemically active sites only
ΔH_{ads}	0.5–5 kcal/g·mol (≈ $\Delta H_{condensation}$)	5–100 kcal/g·mol (≈ bond energy)
Temperature	Active at low T; reversible	Active at high T (> 500 °C relevant); often irreversible
Coverage	Multilayer	Monolayer
Measures	**Surface area + pore volume**	**Active sites**

Both are **exothermic** ($\Delta H_{ads} < 0$). The pairing at the bottom is the exam line, in your notes verbatim: *physisorption is non-specific so it counts all the surface; chemisorption needs bond formation so it counts only the catalytic sites.* That sentence alone was worth a mark you did not take.

3.2 Langmuir — assumptions, derivation, limits

Four assumptions (your notes, in order): entire surface equally active; **no interaction** between adsorbed molecules; heat of adsorption constant and independent of coverage θ; **monolayer at most**.

Derivation — six lines, write it out until it is automatic:

$$r_{ads} = kp(1 - \theta) \qquad r_{des} = k'\theta \qquad \text{(8-1), (8-2)}$$

$$\text{at equilibrium: } kp(1 - \theta) = k'\theta \;\Rightarrow\; \theta(k' + kp) = kp$$

$$\theta = kp/(k' + kp), \quad \text{with } K = k/k' \;\Rightarrow\; \boldsymbol{\theta = Kp/(1 + Kp) = v/v_m} \qquad \text{(8-3)}$$

The two limits (a standard one-mark question):

- `Kp << 1` → θ ≈ Kp — linear, Henry's law region, apparent **first order**.
- `Kp >> 1` → θ → 1 — saturated monolayer, apparent **zero order**.

Linearisation — the class form: invert $\theta = v/v_m$ to get

$$1/v = 1/(K\,v_m)\cdot(1/p) + 1/v_m \quad \rightarrow \text{ plot } 1/v \text{ vs } 1/p\text{: intercept } 1/v_m,\ \text{slope } 1/(Kv_m)$$ class

The loading form used in Tutorial 1 is the same algebra: $p/q = 1/(q_{max}K) + p/q_{max}$ — plot p/q vs p, **slope = $1/q_{max}$, intercept = $1/(q_{max}K)$**, so **K = slope/intercept**. Either is accepted; state which you used.

3.3 The other two isotherms — class only, not in your study PDF

- **Temkin.** Drops the constant-ΔH assumption: $\Delta H_{ads} \propto 1/\theta$, giving $\theta = k_1 \ln(k_2 p)$, both constants at fixed T.
- **Freundlich.** Logarithmic fall of heat with coverage, $\Delta H_{ads}(\theta) = \Delta H_0 \ln\theta$, giving $\theta = C\,p^{1/n}$ with n > 1 — empirical, no saturation limit.

One-line discriminator: **Langmuir assumes uniform surface; Temkin and Freundlich are the two ways of admitting it is not.**

3.4 IUPAC isotherm types — class only

Type	Shape	Means
I	Rises then flattens to a plateau	Monolayer — follows the Langmuir pattern; microporous
II	Knee, then long rise, then sweeps up	Multilayer: first layer chemisorbed, rest physisorbed. **This is the BET case**
III	Convex, no knee	Weak adsorbate–adsorbent interaction
IV	Type II with a plateau and hysteresis	**Capillary condensation** — mesopores
V	Type III plus a step	Weak adsorption **+** capillary condensation

x-axis is always **p/p_0**, where p_0 is the vapour pressure of the adsorbate.

3.5 BET — plot to surface area

BET extends Langmuir to multilayers. Start from $v/v_m = Kp/(1+Kp)$ and you get the form in your notes:

$$p\,/\,[v(p_0 - p)] = 1/(v_m c) + [(c-1)/(c\,v_m)]\,(p/p_0) \tag{8-9}$$

Plot the left side against p/p_0 — a straight line whose two unknowns are v_m and c:

$$I = 1/(v_m c) \quad\cdot\quad s = (c-1)/(v_m c) \quad\Rightarrow\quad \mathbf{v_m = 1/(I + s)} \quad\cdot\quad \mathbf{c = s/I + 1} \qquad \text{(8-10) to (8-12)}$$

Then convert monolayer volume to area — molecules first, then area per molecule:

$$S_g = (N_0 v_m / V_{STP}) \cdot \alpha / m \quad N_0 = 6.02 \times 10^{23},\ V_{STP} = 22{,}400\ \text{cm}^3/\text{mol} \tag{8-13}$$

$$N_2 \text{ at } -195.8\ °\text{C: } \alpha = 16.2\ \text{Å}^2 \Rightarrow \mathbf{S_g = 4.35 \times 10^4 \cdot v_m} \quad (\text{cm}^3/\text{g in, cm}^2/\text{g out}) \tag{8-15}$$

Meaning of c: $c = \exp[(E_1 - E_L)/RT]$, where E_1 is the heat of adsorption of the first layer and E_L the heat of vaporisation. Large c ⇒ the first layer binds much more strongly than later layers condense ⇒ the monolayer completes cleanly ⇒ **sharp knee, well-defined v_m**. As c → 1 the knee disappears and v_m becomes meaningless.

Why nitrogen: cheap, and its α is well established. The apparatus in your lecture slide: sample degassing → test station, carrier gas N_2 + He, adsorbate N_2.

Pores and pore-volume distribution

THE NUMBER YOU GUESSED WRONG

Your answer was “micro < 5, macro > 20”. **Two different conventions exist and you should know which is whose:**

- **IUPAC, and your class notes:** micropores **< 2 nm**; mesopores **2–50 nm**; macropores **> 50 nm** (diameter). **Use this one** — it is what was taught.
- **J. M. Smith (Table 8-3):** a single boundary at **100 Å radius**; below it micropores, above it macropores.

Either way the physics is the same, and it is the sentence in your notes: **“micropores are inside the catalyst particle, macropores are the pores outside/between particles”** — so the **micropores hold essentially all the surface area**, and the macropores are the transport highways feeding them.

4.1 Mercury intrusion — Ritter and Drake

Mercury does **not wet** the surface, so it must be pushed in; the bigger the pressure, the smaller the pore it reaches.

$$a = -2\sigma \cos\theta / p \qquad \theta \approx 140° \text{ (mercury, non-wetting)} \Rightarrow \cos\theta < 0 \Rightarrow a > 0 \tag{8-18}$$

$$\text{practical form:} \quad a\ (\text{Å}) = 8.75 \times 10^5 / p\ (\text{psi}) \tag{8-19}$$

Sanity numbers: 100 Å needs ~10,000 psi; below that you need special high-pressure apparatus. So **porosimetry does the macropores**, and small pores are left to nitrogen.

4.2 Nitrogen desorption — know it exists, formulas are low priority

You wrote “might not be in the syllabus” on C5. Half right: the **lecture plan does list JMS 8-7**, so it is formally in scope, but **your class notes stop at mercury intrusion** and never write the Kelvin or Halsey equations. So: know the *idea*, skip the memorisation unless everything else is solid.

- Nitrogen **wets** the surface ($\theta = 0°$) and condenses in small pores. As you lower the pressure, condensate evaporates out of progressively larger pores — that is the distribution.
- Kelvin equation gives the radius from p/p_0; a Halsey correction δ accounts for the adsorbed layer already lining the wall, so the true radius is $a = (a - \delta) + \delta$.
- For N_2 at −195.8 °C, in Ångstroms: $a - \delta = 9.52[\log(p_0/p)]^{-1}$ and $\delta = 7.34[\ln(p_0/p)]^{-1/3}$. Watch **log vs ln** — that is the trap.
- Range split: nitrogen desorption cannot see above ~200 Å; mercury cannot easily reach below ~100 Å. **The two methods are combined** to cover a bidisperse catalyst.

4.3 The pelleting-pressure result (B6)

Pressing harder collapses the **macropores** (space *between* particles): 1.08 → 0.265 cm^3/g. It does not crush the particles, so **micropore volume barely changes** and, since all the area is in micropores, S_g stays put (389 → 381 m^2/g). Reactor consequence: **the active sites survive but the transport highways feeding them are narrowed** — intraparticle diffusion resistance rises and the global rate can fall even though intrinsic activity per gram is unchanged.

Densities and void volume

The lecture slide gives **five** densities. Learn the denominators — that is the entire content:

Name	Mass of …	divided by volume …
True / solid density ρ_s	solids	actual solid volume, **excluding all voids** — what helium measures
Skeletal density	solids	excluding interparticle voids and **open** pores, but **including closed** pores
Pellet / particle density ρ_p	the pellet	volume of the pellet, pores included — what mercury measures
Bulk density ρ_B	solids	**loosely** packed bed volume — includes interparticle void
TAP density	solids	**densely** (tapped) packed volume

Pellet porosity ε_p = void volume of pellet / total volume of pellet.

5.1 The helium–mercury experiment — the whole recipe

- **Helium** penetrates the pores, so it displaces only the skeleton → $\rho_s = m / V_{He}$.
- **Mercury** will not wet and cannot enter pores at atmospheric pressure, so it displaces the whole particle envelope → $\rho_p = m / V_{Hg}$.
- The **difference is the pore volume**: $V_g = (V_{Hg} - V_{He}) / m$.

$$\varepsilon_p = V_g\rho_s / (V_g\rho_s + 1) = V_g\rho_p \quad \text{— two routes, same answer} \qquad (8\text{-}16),\ (8\text{-}17)$$

Wheeler average pore radius: $\bar{a} = 2V_g / S_g$ — **monodisperse pores only** (8-26)

Numbers worth carrying: typical $\varepsilon_p \approx 0.5$; a fixed bed is roughly **30% pore, 30% solid, 40% interparticle void**.

The trap in $\bar{a}$: if the catalyst is **bidisperse**, the “average” radius lands in the empty valley between the micropore and macropore peaks and describes nothing real. Apply it to the micropore volume alone if you must.

Catalyst types, preparation, promoters

6.1 Classification by electron mobility

Class	Examples	Good for
Conductor (metal)	Fe, Cu, Pt, Pd, Rh, Ni	Hydrogenation, dehydrogenation
Semiconductor (oxide)	NiO, ZnO, Cu_2O	Oxidation, desulfurization
Insulator	Al_2O_3, SiO_2, silica–alumina	Dehydration, cracking, isomerization (acid sites)

6.2 The anatomy of a real catalyst

Your notes draw it as one picture: **active agent** (Pt / Pd / Ag / Ni) dispersed on a **support or carrier** (SiO_2, Al_2O_3, SiC) with **promoters or inhibitors** (K_2O, Cl) mixed in. Example: **V_2O_5 on SiO_2** for $SO_2 \rightarrow SO_3$ in sulfuric-acid manufacture. Supported vs unsupported: bulk iron for ammonia synthesis is the classic **unsupported** catalyst.

- **Carrier / support** — a high-area, mechanically and thermally stable solid that carries a small amount of expensive active agent. **What it buys: maximum active area per gram of active metal, plus resistance to sintering.** That was A10.
- **Promoter** — added **during preparation**, little activity itself, improves activity, selectivity or **stability**. Al_2O_3/K_2O in ammonia iron stops loss of area by sintering.
- **Inhibitor** — the anti-promoter, added deliberately to **suppress an unwanted reaction** and so raise selectivity. Halogen compounds on Ag for ethylene oxide.
- **Poison** — arrives **during operation**, in the feed or from the reaction, and destroys activity (coke, sulfur on Ni/Cu/Pt).

> CORRECT YOUR B7 ANSWER
>
> You wrote that a promoter “increases the consumption of catalyst”. A promoter **does not get consumed and is not consumed faster** — nothing in catalysis is consumed. It **modifies** the active agent: more activity, better selectivity, or longer life. Keep the three words separate by **when they arrive**: promoter and inhibitor go in **at the factory**, poison arrives **in the plant**.

6.3 Manufacture — the flowsheet from your notes

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Precipitation / gel → decantation or filtration → washing → drying
→ crushing → calcination → impregnation → activation
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class

- **Precipitation example:** $Al(NO_3)_3 + NH_4OH \rightarrow Al(OH)_3 + NH_4NO_3$.

- **Sol–gel:** hydrolysis of a precursor salt gives a liquid suspension (nm–μm), condensation turns it into a gel.
- **Impregnation, wet vs dry: wet** uses excess precursor solution and works by diffusion + adsorption; **dry** (incipient wetness) uses exactly the pore volume of solution and works by capillary flow.
- **Activation last, in situ:** reduce NiO to Ni **in the reactor** by passing hydrogen, so air and other gases never touch the reactive nickel.

Dispersion beats loading: 5 wt% NiO can be less active than 0.5 wt% because the concentrated solution deposits **bigger nickel particles**. Building 5.0 wt% as **ten separate 0.5 wt% depositions was 11× more active** than adding it at once. The property that decides activity is **dispersion** — active area per gram of metal — not weight percent.

Numerical drills with the answers

Do these on paper tonight, closed book. Each is two or three lines. The answers are the ones from your test.

C1 External area of spheres — and the unit trap

Sphere: area/volume = $6/d_p$; per gram divide by density: $S_g = 6/(\rho_p d_p)$. You had both. The marks went on units.

- d_p = 60 μm = **6×10^{-3} cm**. Then $6/d_p$ = 1000 cm²/cm³. Converting: 1 cm²/cm³ = 100 m²/m³, so **1×10^5 m²/m³**.
- $S_g = 6/(1.4 \times 6{\times}10^{-3})$ = 714 cm²/g = **0.071 m²/g**.
- Inverted for a target area: $d_p = 6/(\rho_p S_g)$ — you wrote 250/6×1.4, which is upside down. With S_g = 250 m²/g = $250{\times}10^4$ cm²/g: **$d_p = 1.7 \times 10^{-6}$ cm ≈ 171 Å**.
- The point of (d): that particle is molecular. **Real catalyst area is internal, in the pores** — external area is negligible.

C2 Langmuir fit

Linearise **before** you regress: p/q vs p. From the data, slope 0.200, intercept 0.100 ⇒ **q_{max} = 1/slope = 5.0 mmol/g**, **K = slope/intercept = 2.0 atm^{-1}**. Yes, your calculator's regression does this in seconds — but **write the linearised form and the slope/intercept on the paper**, because that is where the marks live, not in the final number.

C3 BET plot → S_g

- $v_m = 1/(I+s) = 1/(15.3{\times}10^{-3})$ = 65.4 cm³ total; **divide by sample mass**: 65.4/0.85 = **76.9 cm³(STP)/g**. (Forgetting the mass is the classic error.)
- Molecules per gram = (76.9/22,400) × $6.02{\times}10^{23}$ = $2.07{\times}10^{21}$; times $16.2{\times}10^{-16}$ cm² = $3.35{\times}10^6$ cm²/g = **335 m²/g**. Shortcut check: $4.35{\times}10^4$ × 76.9 = same.
- c = s/I + 1 = **51** — large, so a sharp knee and a trustworthy v_m.
- With O_2 (α = 14.2 Å²): same v_m, area scales with α ⇒ 335 × 14.2/16.2 = **293 m²/g**.

C4 He–Hg densities

50.0 g, 20.0 cm³ He, 42.0 cm³ Hg, S_g = 200 m²/g:

- ρ_s = 50/20 = **2.50 g/cm³** · ρ_p = 50/42 = **1.19 g/cm³** · V_g = (42−20)/50 = **0.44 cm³/g**
- ε_p = (0.44×2.50)/(0.44×2.50 + 1) = 1.10/2.10 = **0.524**; check 0.44×1.19 = **0.524** ✓
- $\bar{a} = 2(0.44)/(200{\times}10^4) = 4.4{\times}10^{-7}$ cm = **44 Å**

- $p = 8.75\times10^5/44 \approx$ **2×10^4 psi** — beyond an ordinary porosimeter, so **nitrogen desorption** is the right instrument for this pore size.

C5 Kelvin–Halsey point — only if time permits

$p/p_0 = 0.60 \Rightarrow p_0/p = 1.667$. $a - \delta = 9.52/\log(1.667) = 9.52/0.2218 =$ **42.9 Å**; $\delta = 7.34/(\ln 1.667)^{1/3} = 7.34/0.799 =$ **9.2 Å**; **a ≈ 52 Å**. The correction is 21% of the answer, so it matters.

The closed-book drill

Blank paper, no notes, 25 minutes. If you can produce all nine, you are at roughly 40/56 on the same paper. Then re-take the practice test — it clears with the "Clear all answers" button.

1. The seven steps in order, each marked P or C, with 2 and 6 struck out for a nonporous pellet.
2. Eq. (7-1) to (7-4), the four lines, ending in the resistances-in-series form. Then sketch the three concentration profiles.
3. The Langmuir derivation, from r_{ads} = kp(1−θ) to θ = Kp/(1+Kp), plus both limits and the order in each.
4. The BET equation, and from a plot: v_m = 1/(I+s), c = s/I+1, $S_g = (N_0 v_m/22{,}400)\alpha/m$, and the 4.35×10^4 shortcut.
5. The five density definitions by their denominators, then the He–Hg recipe and both porosity routes.
6. Pore sizes: 2 nm / 50 nm, micropores inside, macropores between, area lives in the micropores.
7. a(Å) = 8.75×10^5/p(psi), θ_{Hg} = 140°, θ_{N2} = 0°.
8. Metal / semiconductor / insulator with one reaction each; promoter / inhibitor / poison in one line each.
9. The preparation flowsheet, eight boxes, and what wet vs dry impregnation means.

NUMBERS TO CARRY IN YOUR HEAD

- N_2 at −195.8 °C: α = 16.2 Å², ρ_{liq} = 0.808 g/cm³, $S_g = 4.35\times10^4\ v_m$. O_2 at −183 °C: α = 14.2 Å².
- $N_0 = 6.02\times10^{23}$; V_{STP} = 22,400 cm³/mol.
- ΔH_{ads}: physical 0.5–5 kcal/mol, chemical 5–100 kcal/mol. Both exothermic.
- Pores: <2 nm micro, 2–50 nm meso, >50 nm macro (class); 100 Å single boundary (Smith).
- θ: mercury 140°, nitrogen 0°. 100 Å needs ~10,000 psi of mercury.
- ε_p ≈ 0.5; fixed bed ≈ 30% pore / 30% solid / 40% interparticle.
- FCC context from the slides: reactor 480–550 °C, regenerator 650–715 °C, 1.5–3 atm, catalyst/oil 5.5, residence 2–5 s, Y-zeolite 50–100 μm.

Scored against your submission of 25 August 2026 (9.8 minutes, 6/56). Sources, in the order this sheet trusts them: your class notes (21 pages, 22 Aug), the lecture slide deck `CRE-II_Q1`, and J. M. Smith §7-1 to §7-2 (pp. 273–281) and §8-3 to §8-10 (pp. 287–324). Where the two disagree — the pore-size convention — the class notes win and both are given. §8-11 poisons and Chapter 9 remain out of scope.